

Generation Interconnection System Impact Study Report

For Queue Project Q20-1438

West Lafayette Cumberland Avenue 100 kV
December 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1438. The Interconnection Customer proposes a 180.1 MW solar plus storage facility interconnecting at the West Lafayette Cumberland Avenue 100 kV substation in the Ohio area. The request was received on March 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 410 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$79,051,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SHINROCK / E FRANKLIN 100.0 kV
Contingency: BEAVR / TREGO 500.0 kV L
Loading: 241.5% (Rating: 121 MVA)
DFAX: 57.4%
Pre-project loading: 155.9%
2. Facility: SHINROCK / AVERY 100.0 kV
Contingency: BEAVER / TREGO 500.0 kV
Loading: 196.7% (Rating: 117 MVA)
DFAX: 72.0%
Pre-project loading: 86.1%
3. Facility: AVERY SUBSTATION / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV
Contingency: BEAVER / TREGO 500.0 kV
Loading: 196.4% (Rating: 127 MVA)
DFAX: 72.0%
Pre-project loading: 93.9%
4. Facility: JOHN P MADGETT SUBSTATION / DOLLARHIDE 100.0 kV line
Contingency: BEAVER SUBSTATION / EUFAULA 500.0 kV line
Loading: 167.7% (Rating: 128 MVA)
DFAX: 5.3%
Pre-project loading: 160.3%
5. Facility: ROCK RAPIDS / IDYLVILD 100.0 kV
Contingency: TREGO / BEAVER 500.0 kV
Loading: 161.3% (Rating: 159 MVA)
DFAX: 27.0%
Pre-project loading: 130.7%
6. Facility: IDYLVILD / RIPP 100.0 kV
Contingency: TREGO / BEAVER 500.0 kV line
Loading: 149.1% (Rating: 190 MVA)

DFAX: 27.0%

Pre-project loading: 123.6%

7. Facility: TIMBERLINE / CASTALIA 100.0 kV

Contingency: AVERY SUBSTATION / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV

Loading: 141.1% (Rating: 163 MVA)

DFAX: 58.4%

Pre-project loading: 76.7%

8. Facility: AXTEL / RIPP 100.0 kV

Contingency: TREGO / BEAVER 500.0 kV L

Loading: 140.7% (Rating: 220 MVA)

DFAX: 27.0%

Pre-project loading: 118.6%

9. Facility: WHISKEYTOWN / E FRANKLIN 100.0 kV L

Contingency: BEVER SUBSTATION / TREGO 500.0 kV

Loading: 132.5% (Rating: 260 MVA)

DFAX: 27.0%

Pre-project loading: 113.8%

10. Facility: POTTER / CASTALIA 100.0 kV Line

Contingency: AVERY / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV

Loading: 127.8% (Rating: 128 MVA)

DFAX: 40.9%

Pre-project loading: 70.1%

11. Facility: FLAT ROCK / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV

Contingency: AVERY / WEST LAFAYETTE CUMBERLAND AVENUE 100.0 kV

Loading: 122.3% (Rating: 135 MVA)

DFAX: 100.0%

Pre-project loading: 10.9%

12. Facility: FLAT ROCK / ENERGY SHELBY COUNTY 100.0 kV

Contingency: W LAFAYETTE CUMBERLAND AVENUE SUBSTATION / AVERY 100.0 kV Line

Loading: 111.6% (Rating: 138 MVA)

DFAX: 100.0%

Pre-project loading: 19.4%

13. Facility: POWERSVILLE RD / ENERGY SHELBY COUNTY 100.0 kV

Contingency: AVERY / W LAFAYETTE CUMBERLAND AVENUE SUBSTATION 100.0 kV

Loading: 100.6% (Rating: 140 MVA)

DFAX: 100.0%

Pre-project loading: 27.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EAST FRANKLIN / SHINROCK 100.0 kV	Rebuild line with higher-capacity conductor	\$5,579,000	100% (\$5,579,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHINROCK / AVERY 100.0 kV	Rebuild line with higher-capacity conductor	\$2,465,000	100% (\$2,465,000)
AVERY / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,706,000	100% (\$2,706,000)
JOHN P MADGETT SUBSTATION / DOLLARHIDE 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,571,000	100% (\$1,571,000)
ROCK RAPIDS / IDYLVILD 100.0 kV	Rebuild line with higher-capacity conductor	\$2,960,000	100% (\$2,960,000)
IDYLVILD / RIPP 100.0 kV	Rebuild line with higher-capacity conductor	\$5,552,000	100% (\$5,552,000)
TIMBERLINE / CASTALIA 100.0 kV	Rebuild line with higher-capacity conductor	\$10,666,000	100% (\$10,666,000)
AXTEL / RIPP 100.0 kV	Rebuild line with higher-capacity conductor	\$2,469,000	100% (\$2,469,000)
WHISKEYTOWN / E FRANKLIN 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,214,000	100% (\$6,214,000)
POTTER / CASTALIA 100.0 kV Line	Rebuild line with higher-capacity conductor	\$4,684,000	100% (\$4,684,000)
FLAT ROCK / W LAFAYETTE CUMBERLAND AVENUE 100.0 kV	Rebuild line with higher-capacity conductor	\$22,305,000	100% (\$22,305,000)
FLAT ROCK / ENERGY SHELBY COUNTY 100.0 kV	Rebuild line with higher-capacity conductor	\$8,851,000	100% (\$8,851,000)
POWERSVILLE RD / ENERGY SHELBY COUNTY 100.0 kV	Reconductor line	\$3,029,000	100% (\$3,029,000)
	Total allocated to project		\$79,051,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.